M|P|W

OCR
Oxford Cambridge and RSA

Physics A (H156, H556)

Chapter 7 TA_2025
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **38**.
- The total number of marks may take into account some 'either/or' question choices.

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1. Which physical quantity has the same base units as energy?

A. moment
B. momentum
C. force
D. pressure

Your answer ☐

[1]

2. A ball, initially at rest, is struck by a hockey stick. It leaves the hockey stick at speed *v*.

Which quantity, together with the mass of the ball, can be used to determine *v*?

A The time of the impact.

B The weight of the hockey stick.

C The impulse of the force.

D The final momentum of the hockey stick.

Your answer ☐ **[1]**

3. A puck of mass 0.16 kg is sliding on ice with a constant velocity of 11.0 m s^{-1}. A hockey stick exerts a force on the puck, for a short period of time, in the **opposite** direction to the velocity of the puck. The momentum of the puck changes by 2.0 kg m s^{-1}.

Ignore friction.

What is the speed of the puck when it leaves the hockey stick?

A 1.5 m s^{-1}

B 3.8 m s^{-1}

C 12.5 m s^{-1}

D 23.5 m s^{-1}

Your answer ☐ **[1]**

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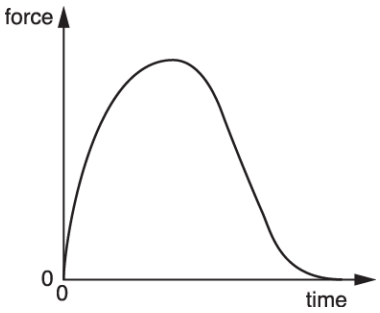
4. The graph shows the resultant force on a football as it is kicked.



Which of the following graphs relating to this kick would have the same shape as the graph above?

A acceleration of the ball against time
B kinetic energy of the ball against time
C momentum of the ball against time
D velocity of the ball against time

Your answer ☐

[1]

5. An object **P** is travelling to the right with a momentum of 40 kg m s^{-1}. It collides with another object **Q** travelling to the left along the same path.

The final momentum of **P** is 10 kg m s^{-1} to the right.

What is the change in the momentum of **Q**?

A 0 $kgms^{-1}$
B 10 $kgms^{-1}$
C 30 $kgms^{-1}$
D 50 $kgms^{-1}$

Your answer ☐ **[1]**

6. A javelin thrower exerts a force of 100 N on a javelin for a time of 0.30 s.
The javelin has a mass of 0.80 kg.

What is the rate of change of the momentum of the javelin?

A 24 kg ms^{-2}
B 30 kg ms^{-2}
C 100 kg ms^{-2}
D 125 kg ms^{-2}

Your answer ☐ **[1]**

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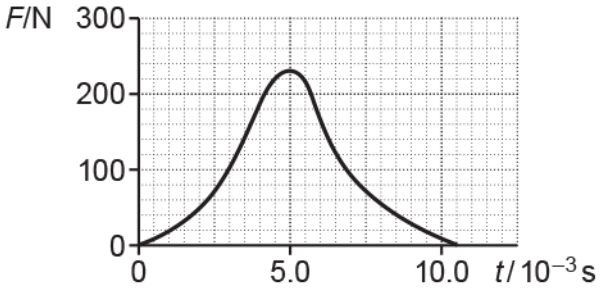
7. The variation with time t of the force *F* acting on a ball is shown below.

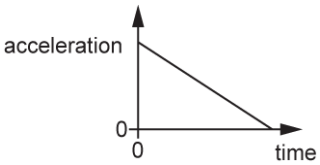


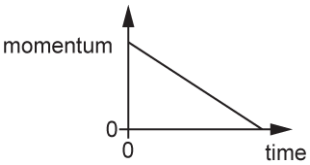
Which statement is **not** correct?

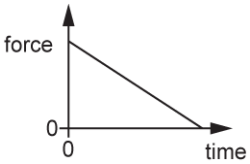
A The area under the graph is equal to the work done by the force *F*.
B The ball has maximum acceleration at $t = 5.0 \times 10^{-3}$ s.
C The area under the graph is equal to impulse.
D The area under the graph has units kg m s^{-1}.

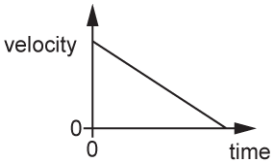
Your answer ☐ **[1]**

8. Which one of the graphs below can be used to determine the impulse of a force from the area under the graph?

A


B


C


D


Your answer ☐ **[1]**

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9. A trolley **M** collides head-on with a trolley **L**. The mass of trolley **M** is greater than the mass of trolley **L**. The trolleys join together after the collision.

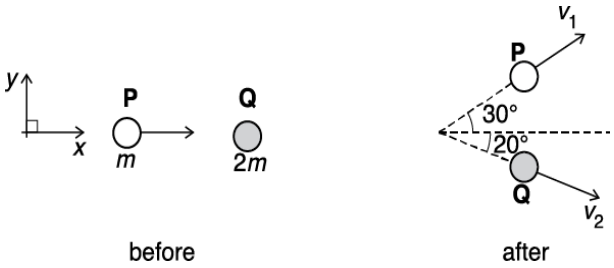
Which statement is correct?

A The momentum of each trolley is conserved.
B Trolley **M** experiences a greater force than trolley **L** during the collision.
C The total force acting on the two-trolley system during the collision is zero.
D Kinetic energy is conserved.

Your answer ☐

[1]

10. A ball **P** of mass m has a velocity in the positive x-direction. It makes a collision with a stationary ball **Q** of mass $2m$. After the collision, the ball **P** has velocity v_1, ball **Q** has velocity v_2 and the balls travel in the directions shown in the diagram below.



After the collision, the total momentum of the balls in the x-direction is p_x and the total momentum in the y-direction is p_y.

Which row is correct for p_x and p_y?

	p_x	p_y
A	$2mv_2 \cos 20° + mv_1 \cos 30°$	0
B	$2mv_2 \sin 20° + mv_1 \sin 30°$	0
C	$2mv_2 \cos 20° + mv_1 \cos 30°$	$2mv_2 \sin 30° + mv_1 \sin 20°$
D	$2mv_2 \sin 20° + mv_1 \sin 30°$	$2mv_2 \cos 30° + mv_1 \cos 20°$

Your answer ☐

[1]

11(a). A sports manufacturer is testing the quality of one of their footballs.

Fig. 3.1 shows how the force F applied to a football varies with time t whilst it is being kicked horizontally. The ball is initially at rest.

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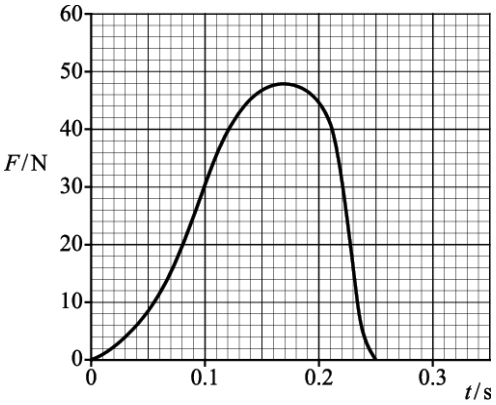

Fig. 3.1

i. Use the graph to find:

1. the maximum force applied to the ball ..... N

2. the time the boot is in contact with the ball ..... s.

[1]

ii. The mean force multiplied by the time of contact is called the impulse delivered to the ball. The impulse delivered to the ball is about 6.5 N s.

Explain how you would use the graph to show that the impulse has this value.

---**[2]**

(b). The mass of the ball is 0.60 kg. Use your answers in **(a)** to calculate

i. the maximum acceleration of the ball

acceleration = ... m s^{-2} **[2]**

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ii. the final speed of the ball.

speed = ... m s^{-1} **[2]**

(c). The ball hits a wall with a speed of 11 m s^{-1}. It rebounds from the wall along its initial path with a speed of 6.0 m s^{-1}. The impact lasts for 0.18 s.

Calculate the mean force exerted by the ball on the wall.

mean force = ... N **[3]**

12(a). State *Newton's second law.*

.....

.....

..... **[1]**

(b). A hydrogen atom travelling at 500 m s^{-1} makes a head-on collision with a stationary carbon atom. The collision is perfectly elastic. After the collision the hydrogen atom bounces back with a speed of 420 m s^{-1}. Fig. 24.2 shows the atoms before and after the collision.

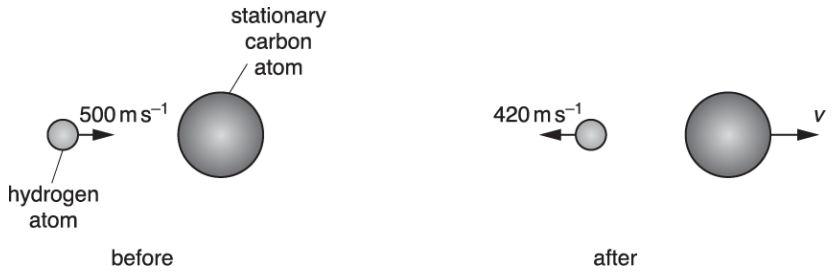

Fig. 24.2

The mass of the hydrogen atom is 1.7×10^{-27} kg and the mass of the carbon atom is 2.0×10^{-26} kg.
Calculate the speed *v* of the carbon atom after the collision.

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v = ... m s^{-1} **[3]**

13.

A massive ball is released from rest above the ground.
According to a student, the principle of conservation of momentum is violated because the ball gains momentum as it falls.
Explain why the student's observation is incomplete and discuss how momentum is conserved in this situation.

---**[2]**

14.

Fig. 5.1 shows a tennis ball before and after bouncing on the ground.

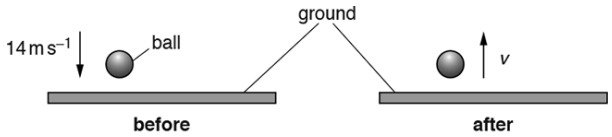

Fig. 5.1

The mass of the tennis ball is 0.062 kg. The tennis ball is slightly warmer after its collision with the ground.

i. The tennis ball hits the ground at a speed of 14 m s^{-1}.
Calculate the momentum p of the tennis ball as it hits the ground.

p = ---------- Ns **[1]**

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The force acting on the ball during collision with the ground is *F*.
Fig. 5.2 shows a graph of force *F* acting on the tennis ball against time *t*.

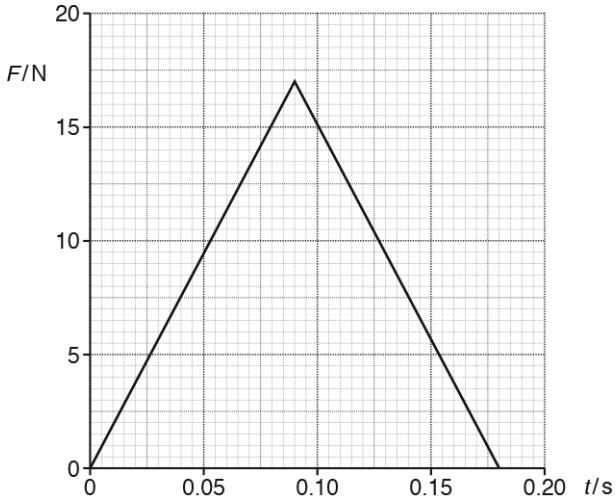

Fig. 5.2

The tennis ball is in contact with the ground for 0.18 s.

ii. Determine the speed *v* of the tennis ball as it leaves the ground.

v = ---------- m s^{-1} **[3]**

iii. State what is meant by an elastic collision and explain how your answer to **(ii)** shows that this collision is **not** elastic.

---**[2]**

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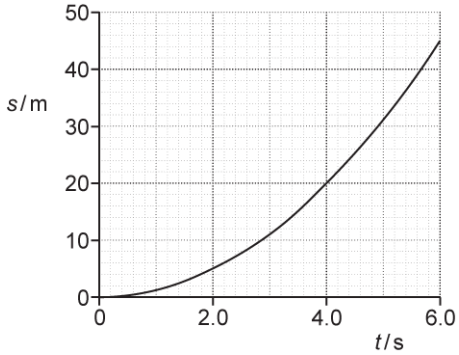
15. A car starts from rest at time $t = 0$.

The car travels in a straight line with a **constant** acceleration.

The displacement s against time t graph for this car is shown below.



i. Use the graph to show that the speed of the car at $t = 4.0$s is $10 ms^{-1}$.

[3]

ii. The mass of the car is 1200kg.

Calculate the rate of change of momentum of the car from $t = 0$ to $t = 4.0$ s. Include an appropriate unit for your answer.

rate of change of momentum = ... unit ..... **[3]**

END OF QUESTION PAPER

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Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			A	1	
			Total	**1**	
2			**C**	**1**	
			Total	**1**	
3			**A**	1	**Examiner's Comments** This question was more challenging still. In this question, candidates often forgot that the impulse provided by the hockey stick is in the opposite direction to the momentum of the puck, again giving option A as the correct answer.
			Total	**1**	
4			A	1	**Examiner's Comments** All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions.
			Total	**1**	
5			**C**	1	
			Total	**1**	
6			**C**	1	**Examiner's Comments** There was a significant amount of working shown by candidates across the ability spectrum. Most of the time, the answers put down were the distractors. This question is about Newton's second law of motion. The force on the javelin is equal to the rate of change of its momentum. Therefore, the answer is simply **C**. The kg m s^{-2} is equivalent to the newton N.
			Total	**1**	
7			**A**	1	**Examiner's Comments** This question was based on a force-time graph for a ball. All

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					the information that can be extracted from the graph. The majority of the candidates inserted A in the answer box, and secured 1 mark. All the other statements are correct. The statement B is correct, because the maximum acceleration of the ball is proportional to the maximum force. The area under a force-time graph is impulse, so statement C is correct. The area under the graph is also equal to change in momentum, therefore the area under the graph does have the unit's kg m s^{-1}. So, statement D is also correct. Statement A is incorrect. Work done is the area under a force-distance graph, but here the horizontal axis has **time**, and not **distance**.
			Total	**1**	
8			C	1	
			Total	**1**	
9			C	1	**Examiner's Comments** All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions. The correct key was **C**: *The total force acting on the two-trolley system during the collision is zero.* The most frequent distractor was **A**: *The momentum of each trolley* is *conserved*. The term '*each*' did not register with most candidates. It is the total momentum of the two trolleys that is conserved.
			Total	**1**	
10			A	1	
			Total	**1**	
11	a	i	48 (N); 0.25 (s)	A1	both values correct; no tolerance
		ii	estimate area under graph by counting squares **or**	B1 B1	second mark awarded for some detail about how to estimate area.
		ii	method of estimating mean F multiplied by time of contact (0.25 s)		e.g. areas above and below mean under curve are equal.
	b	i	(F = ma) a = 48/0.6	C1	**ecf a(i)**

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		i	a = 80 (m s^{-2})	A1	
		ii	(Ft = mv) v = 6.5/0.6	C1	**ecf a(i)**
		ii	v = 11 (m s^{-1})	A1	allow 10.8
	c		Ft = mv + mu	C1	
			Ft = 0.6 (6 + 11) = 10(.2)	C1	
			F = 10/0.18 = 57 (N)	A1	award 1 mark out of 3 for mv − mu; = 0.6 (11 − 6) = 3 giving 17 (N).
			Total	**10**	
12	a		(Resultant) force is (directly) proportional / equal to the rate of change of momentum	B1	**Not** force = mass × acceleration **Not** ' force ∝ change in momentum over time' **Examiner's Comments** A few candidates answered this question well demonstrating their knowledge of this fundamental law of physics. The most frequent incorrect answer was '*Force is equal to mass multiplied by acceleration*' rather than '*Force is proportional to the rate of change of momentum*'.
	b		*Method 1: Momentum is conserved*		
			$1.7 \times 10^{-27} \times 500$ or $1.7 \times 10^{-27} \times (-)\ 420$ or $2.0 \times 10^{-26} \times v$	C1	
			$1.7 \times 10^{-27} \times 500 = 1.7 \times 10^{-27} \times -420 + 2.0 \times 10^{-26} \times v$	C1	
			v = 78 (m s^{-1})	A1	**Allow** 1 mark for 6.8 (m s^{-1}); + 420 used instead of − 420
			Method 2: Kinetic energy is conserved		
			$\frac{1}{2} \times 1.7 \times 10^{-27} \times 500^2$ or $\frac{1}{2} \times 1.7 \times 10^{-27} \times 420^2$ or $\frac{1}{2} \times 2.0 \times 10^{-26} \times v^2$	C1	**Allow** full credit for correct use of 'velocity of approach = −velocity of recession', e.g:
			$\frac{1}{2} \times 1.7 \times 10^{-27} \times 500^2 = \frac{1}{2} \times 1.7 \times 10^{-27} \times 420^2 + \frac{1}{2} \times 2.0 \times 10^{-26} \times v^2$	C1	'speed' of approach = (−) 'speed' of recession **C1** 500 = v + 420 **C1** v = 80 (m s^{-1}) **A1**
			v = 79 (m s^{-1})	A1	**Examiner's Comments** This was a good discriminator with many of the top-end candidates scoring full marks. Most candidates opted to answer the question using the principle of conservation of momentum. A few candidates used ideas of conservation of kinetic energy for this perfectly elastic collision. It is good to

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					report that most candidates coped well with powers of ten. The most common mistake was to use +420 m s^{-1} for the final velocity of the hydrogen atom, rather than −420 m s^{-1}; this gave the incorrect answer of 6.8 m s^{-1}. A small number of candidates used relative velocities before and after to arrive at an alternative correct answer of 80 m s^{-1}.
			Total	**4**	
13			Earth mentioned (as an integral part of the system) The Earth has (equal and) opposite momentum to the (falling) ball (so momentum is conserved) or The Earth moves upwards / towards the ball (with a tiny speed, so momentum is conserved)	**M1** **A1**	**Not** 'ground' **Allow:** The Earth experiences an upward force (and moves upwards) **Examiner's Comments** This question on the principle of conservation of momentum was not answered well. Only a small number of exceptional candidates recognised that momentum of the Earth-ball system was conserved which meant that the Earth must have momentum equal to that of the falling ball but in the *opposite* direction. Most of the candidates thought either air resistance had a role to play, or that momentum was conserved only when the ball collided with the ground. A small number of candidates thought that the conservation of momentum was intrinsically linked with the principle of conservation of energy. This prompted incorrect physics such as *'momentum transferred to KE and then GPE after impact with the ground'*.
			Total	**2**	
14		i	0.868 or 0.87	**B1**	**Allow** – 0.868 or – 0.87 **<u>Examiner's Comments</u>** This question was well answered. Many candidates helpfully showed their working.
		ii	change in momentum = 0.5 × 17 × 0.18 or 1.53 (N s) Momentum of tennis ball = 1.53 − 0.868 = 0.662 v = 0.662/0.062 = 10.6 or 10.7 (m s^{-1}) = 11 (m s^{-1})	**C1** **C1** **A1**	**ECF** from (b)(i) Or 0.062(v- -14) = 1.53 Or v = 24.7 -14 = 10.7 **Allow** 1 mark for 24.7 (m s^{-1}) or 38.7 (m s^{-1}) **<u>Examiner's Comments</u>** Candidates found this question difficult. Candidates needed

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					to realise the importance of Fig. 5.2 and understand that the area under the graph was equal to the change of momentum of the ball. It is important that candidates show their working so that if the graph is misread they may still obtain credit later. Having determined the change in momentum correctly, candidates then needed to determine the momentum of the ball leaves the ground. **Exemplar 4** $F = \frac{mv}{t}$ $Ft = mv$ Area under the ground is impulse $0.5 \times 0.18 \times 17 N = mv$ $1.53 = 0.062 (v - u)$ $(v-u) = 24.677$ $v = 14$ $u = 10.6717 \approx 10.7$ $v =$ 10.7 ms^{-1} **[3]** This exemplar demonstrates that the candidate understands that the area under the graph is the impulse. The method to calculate the area is shown and although the candidate has not used the answer to (b)(i), the method in terms of the change of velocity is correct and clearly indicated.
		iii	For an elastic collision, kinetic energy/<u>KE</u> is conserved speeds are different (so for the same mass KE is different)	**B1** **B1**	**Allow** speed of approach = speed of separation **Allow** correct calculations of KE for both speeds **Ignore** reference to the ball heating up **<u>Examiner's Comments</u>** Many candidates did not state that the kinetic energy is conserved − often there were general statements about energy being conserved. Some candidates did not relate their answer to part (ii) as required by the question. Some candidates determined the kinetic energy before and after the ball left the ground − candidates could gain full marks with this approach.
			Total	**6**	
15		i	Tangent drawn at t = 4.0 s Attempt at calculating the gradient v calculated from gradient and between 9.50 - 10.50 (m s^{-1})	C1 C1 A1 C1	**Allow** other correct methods **Note** working required for this mark

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			OR s = 20 (m) and $s = ½ at^2$ $20 = ½ a \times 4.0^2$ **or** $a = 2.5$ (m s^{-2}) $v = 2.5 \times 4.0$ **or** $v^2 = 2 \times 2.5 \times 20$ $v = 10$ (m s^{-1})	C1 C1 A0	
		ii	change in momentum = 1200 x 10 or 12000 (kg m s^{-1}) rate of change of momentum = 3000 unit: kg m s^{-2} or N **OR** F = 1200 X 2.5 rate of change of momentum = 3000 unit: kg m s^{-2} or N	C1 A1 B1 C1 A1 B1	**Allow** ECF from **(i)** **Allow** 2850 - 3150 **Allow** newton **Allow** ECF from **(i)** **Allow** newton
			Total	**6**	

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